

## HamishAI

Internal Operations Platform

### Business Analysis Documentation Pack

*Covering lead generation, client onboarding, request handling, client support, site monitoring, billing and management reporting*

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Reference model: HamishAI Business Process Model ([admin.hamishai.org/process](http://admin.hamishai.org/process))

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# 1. Executive Summary

## 1.1 Business Problem

Hamish AI is an Edinburgh-based consultancy that designs and delivers AI-powered automation for small businesses. Prior to this platform, the business ran its own client operations — request handling, task tracking, lead generation, and billing — through informal, manual channels: email, phone calls and memory, with no central system of record. As the client base has grown, this created a real operational risk: requests could be lost or delayed, lead follow-up depended on the operator remembering to chase, invoices were raised and chased manually, and clients had no way to check progress without contacting the operator directly. The business problem is therefore twofold: (a) the operational risk of running a growing client-service business on ad hoc tooling, and (b) a credibility gap — a consultancy that sells AI-driven operational automation to its clients was not, itself, running on one.

## 1.2 Current State

Client requests arrived through unstructured channels with no formal lifecycle, lead prospecting was tracked in a spreadsheet with no automated reminder, invoicing (where it existed at all) was raised manually with no chase mechanism, and website health for client sites was checked reactively — typically only once a client reported a problem. Section 5 (Current State Analysis) documents this in full, mapped against the same process structure used for the target operating model.

## 1.3 Future State

The platform now in operation implements a seven-process operating model — Lead Generation, Client Onboarding, Request Handling & Fulfillment, Client Support, Site Monitoring, Billing, and a Daily Overview reporting layer — each supported by purpose-built automation and, in five specific decision points, by AI. An internal admin console gives the operator a single place to manage clients, requests, tasks, leads and invoices; a client-facing self-service portal gives clients visibility of their own requests, tasks, website health and billing without needing to contact the operator. The complete flow is captured in the HamishAI Business Process Model (the reference diagram at [/admin/process](#)) and every artefact in this pack is traceable back to it.

## 1.4 Expected Benefits

- Reduced administrative overhead per client through automated triage, task creation, calendar scheduling and client notification.
- Faster, more consistent client response times, including automatic resolution of routine, maintenance-covered requests.

- Improved cash flow predictability through automatic invoice status reconciliation and one-click overdue reminders.
- A repeatable new-business pipeline through scheduled lead research, AI-drafted outreach and automatic follow-up flagging.
- Earlier detection of client website issues, ahead of the client noticing and reporting them.
- A scalable operating model that supports growth in client volume without a proportional increase in operator time.

## 1.5 Success Measures

Success is measured through the KPI set defined in Section 26 (Process Metrics), spanning auto-resolution rate, time-to-first-response, lead conversion and follow-up compliance, on-time invoice payment rate, site-issue detection rate and portal adoption. These metrics are drawn directly from the underlying data model documented in Section 15 (Data Requirements), so they can be reported without any additional instrumentation.

# 2. Business Objectives

## 2.1 Strategic Objectives

- Scale client-service capacity without a proportional increase in operator headcount or hours.
- Establish Hamish AI's own operations as a live, working reference case for the AI-automation services it sells to clients.
- Improve cash-flow predictability through automated billing status tracking and payment reminders.
- Build a repeatable, low-effort client-acquisition pipeline (research → outreach → follow-up → onboarding).

## 2.2 Business Outcomes

- Faster request turnaround, including same-day resolution of routine, maintenance-covered requests.
- Reduced incidence of late payment through automatic reminders on overdue invoices.
- Consistent, auditable client communication regardless of which channel a client used to make contact.
- A complete, queryable operational history for every client, request, task, lead and invoice.

## 2.3 Measurable KPIs

Full KPI definitions, targets and data sources are set out in Section 26 (Process Metrics). At a glance, the objectives above map to the following KPI families:

| Objective                                   | KPI Family                                                         |
|---------------------------------------------|--------------------------------------------------------------------|
| Scale without proportional headcount growth | Auto-resolution rate; operator daily-review time                   |
| Live reference case for AI automation       | Portal adoption rate; AI decision accuracy (auto-send correctness) |
| Cash-flow predictability                    | Invoice on-time payment rate; overdue reminder effectiveness       |
| Repeatable acquisition pipeline             | Lead-to-client conversion rate; follow-up compliance               |

## 2.4 Critical Success Factors

- AI triage must classify requests accurately and default conservatively to human review when uncertain.
- Clients must adopt and use the self-service portal rather than reverting to email/phone as their default channel.
- The operator must consistently use the admin console and Overview dashboard as the daily point of control.
- Third-party dependencies (Google OAuth token, Stripe configuration, Supabase availability) must be kept current — see Section 17 (Integration Requirements) and Section 19 (Risk Assessment).

## 3. Scope

### 3.1 In Scope

- Client record management, including maintenance plan and package tracking (Process 2 — Client Onboarding).
- Multi-channel request capture (client portal and monitored email inbox) and AI-assisted triage (Process 3).
- Automatic reply for high-confidence, maintenance-covered requests; clarification flagging for incomplete requests (Process 3).
- Task creation, calendar scheduling and client notification on completion (Process 3).
- Client self-service support via an AI-answered knowledge base (Process 4).
- Scheduled website health monitoring and alerting (Process 5).
- Invoice creation, delivery, payment reconciliation and overdue reminders via a real payment processor (Process 6).
- Weekly automated lead research, AI-drafted outreach and stale-lead follow-up flagging (Process 1).
- A consolidated daily Overview dashboard for the operator (Process 7).
- A visual, maintained Business Process Model and this accompanying documentation pack.
- Passwordless (magic-link) access to the admin console as an alternative to the shared password.

### 3.2 Out of Scope

- Recurring or subscription-based invoicing for maintenance-plan clients (identified as a Phase 2 candidate — see Section 32, Recommendations).
- Automatic, unattended card-on-file payment collection (all payment collection is via a client-initiated hosted payment page).
- Multi-operator or role-based access control within the admin console (single-operator model only).
- Native mobile applications (the portal and admin console are responsive web applications only).
- CRM-grade marketing automation beyond a single AI-drafted outreach message and one follow-up flag.
- Multi-currency billing (GBP only).
- Formally contracted SLA guarantees to clients.

### 3.3 Assumptions

- ASM-01: Hamish remains the sole system operator for the documented scope of this platform.
- ASM-02: All clients are UK-registered small businesses invoiced in GBP.
- ASM-03: Clients are willing and able to use a web-based self-service portal; no dedicated client-side application support is required.

- ASM-04: The payment processor account will be promoted from test/sandbox mode to live mode before any real client is charged.
- ASM-05: The Google Cloud OAuth application will either be reconnected manually every 7 days or be moved out of Testing publishing status.
- ASM-06: Client, request and lead volumes remain within the practical review capacity of a single operator for the documented scope; no formal capacity ceiling is defined in this release.

(Full assumptions log with cross-references: Section 20.)

### 3.4 Constraints

- The hosting platform's entry-level plan restricts scheduled (cron) jobs to a once-daily frequency, so several processes are necessarily batch/daily rather than real-time (inbound email ingestion, site-health checks, the weekly digest).
- The platform depends on the continuous availability of five external SaaS providers (AI, data/auth, payments, email delivery, calendar/inbox).
- Single-operator bandwidth constrains the throughput of Process 3 (Request Handling) regardless of the degree of automation applied to triage.

### 3.5 Dependencies

Full dependency log with ownership: Section 21. In summary, the platform depends on: the AI provider (Anthropic), the data and authentication platform (Supabase), the payment processor (Stripe), the email delivery provider (Resend), the calendar and inbox provider (Google Workspace APIs), and the hosting/scheduling platform (Vercel), plus DNS/domain resolution for hamishai.org.

## 4. Stakeholder Analysis

Because Hamish AI operates as a single-person consultancy, several conventional stakeholder roles — Product Owner, Developer, Support Agent, Account Manager and Finance — are formally consolidated into one Business Owner / Operator role. This consolidation is deliberate and is documented explicitly below rather than left implicit, as it materially affects the RACI Matrix (Section 24) and the single-operator risks identified in Section 19.

| Stakeholder | Role                                                                                                    | Responsibilities                                                                      | Infl. | Int. | Communication Approach                      | Concerns                                                                                            | Success Criteria                                                              |
|-------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------|------|---------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Hamish      | Business Owner / Operator (consolidates Product Owner, Developer, Support, Account Management, Finance) | Owns all seven processes end to end; reviews and actions the Overview dashboard daily | High  | High | Direct, continuous use of the admin console | Time overhead of manual review; cost of AI/API usage; avoiding client-facing errors from automation | Reduced daily admin time; no missed client requests; positive client feedback |

| Stakeholder                                             | Role                                            | Responsibilities                                                                  | Infl.                                              | Int.                | Communication Approach                                          | Concerns                                                                 | Success Criteria                                                                 |
|---------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------|---------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Client (e.g. Craigie & Sons Joinery, Ellis Home Bakery) | External — requester and payer                  | Submits requests; responds to clarification questions; pays invoices              | Medium                                             | High                | Email and client portal                                         | Response speed; billing clarity; trust in AI-handled replies             | Fast turnaround; transparent billing; no surprises                               |
| Prospect                                                | External — outreach recipient, not yet a client | None formal; may respond to outreach                                              | Low                                                | Low (until engaged) | One-way email outreach and a single follow-up                   | Relevance and unsolicited contact                                        | Conversion to Client                                                             |
| AI Triage & Drafting Engine (Anthropic Claude)          | System — automated decision-maker               | Classifies requests; drafts replies, outreach and portal answers; scores leads    | High (shapes client-facing communication directly) | N/A                 | System-prompt governance; every output remains human-reviewable | Mis-classification; incorrect auto-send                                  | Consistent, accurate categorisation; safe default to human review when uncertain |
| Stripe (Payment Processor)                              | System dependency                               | Invoice lifecycle, hosted payment page, payment status events                     | High (billing integrity)                           | N/A                 | API integration and signed webhook                              | Correct webhook delivery; sandbox vs live-mode configuration             | Accurate, timely payment status sync                                             |
| Supabase (Data & Auth Platform)                         | System dependency                               | System of record for every entity; authentication for portal and admin magic-link | High (single point of data failure)                | N/A                 | Service-role API access (server-side only)                      | Availability; Row Level Security integrity                               | Continuous availability; no unauthorised data exposure                           |
| Google Workspace (Gmail & Calendar APIs)                | System dependency                               | Inbound email ingestion; calendar sync for tasks                                  | Medium-High                                        | N/A                 | OAuth2, reconnect flow on /admin/google-setup                   | OAuth token expiry while the app remains in Testing status (7-day cycle) | Uninterrupted daily ingestion and calendar sync                                  |
| Vercel (Hosting & Scheduler)                            | System dependency                               | Hosting and execution of daily scheduled automations                              | High                                               | N/A                 | Deployment pipeline and cron configuration                      | Entry-tier plan's daily-only scheduling constraint                       | Reliable daily job execution                                                     |

| Stakeholder             | Role              | Responsibilities                                            | Infl.  | Int. | Communication Approach      | Concerns                               | Success Criteria                    |
|-------------------------|-------------------|-------------------------------------------------------------|--------|------|-----------------------------|----------------------------------------|-------------------------------------|
| Resend (Email Delivery) | System dependency | Transactional email delivery to clients and internal alerts | Medium | N/A  | Domain-verified sending API | Deliverability and domain verification | Reliable delivery; no spam-flagging |

## 5. Current State Analysis (As-Is)

### 5.1 Existing Process

Before this platform, client requests were received by phone, email or in person with no central log. The operator judged each request individually and replied ad hoc, with no formal record of what had been promised or when. Prospective new business (leads) were tracked in a spreadsheet, researched irregularly, and had no reminder mechanism for re-engagement. Invoicing, where it existed, was raised manually with no automated chase for late payment. Client website health was checked reactively, generally only once a client reported that something appeared broken. There was no client-facing channel through which a client could check progress without contacting the operator directly.

### 5.2 Pain Points, Risks and Business Impact

| Area               | As-Is Behaviour                                      | Pain Point / Risk                                      | Business Impact                                    |
|--------------------|------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|
| Request intake     | Informal — phone, email, in person; no central log   | Requests can be lost, forgotten or duplicated          | Client dissatisfaction; reputational risk          |
| Triage             | Manual judgement applied fresh to every request      | Slow, inconsistent response and prioritisation         | Delayed turnaround; inconsistent client experience |
| Task tracking      | Informal — memory and inbox only, no due dates       | No visibility of outstanding work                      | Risk of missed deadlines and dropped commitments   |
| Lead follow-up     | Manual, spreadsheet-based, no reminder               | Leads go cold and are not re-engaged                   | Lost new-business opportunity                      |
| Invoicing          | Manual, ad hoc, no reminder mechanism                | Delayed invoicing; no chase for late payment           | Cash-flow risk                                     |
| Website monitoring | Reactive only, discovered when the client reports it | Downtime, SSL expiry or broken links found late        | Client trust damage; avoidable escalation          |
| Client visibility  | None — clients must email or call for status         | Perceived unresponsiveness; unnecessary contact volume | Extra admin overhead; weaker client relationship   |

## 6. Future State Analysis (To-Be)

### 6.1 New Process

A single integrated platform now implements the seven-process operating model set out in the HamishAI Business Process Model: Lead Generation, Client Onboarding, Request Handling & Fulfillment, Client Support, Site Monitoring, Billing, and Daily Overview reporting. Every request, regardless of channel, is logged centrally and triaged automatically. High-confidence, maintenance-covered requests are answered without operator involvement; incomplete requests are flagged with specific clarifying questions rather than left for manual review. Genuine work becomes a tracked task with a calendar entry, and the client is notified automatically when it is done.

### 6.2 AI Automation

Five specific decisions are AI-assisted rather than manual: request triage classification, auto-send eligibility, missing-information detection, lead-to-case-study matching for outreach, and knowledge-base question answering. Full detail, including inputs, logic, outputs, confidence handling and human override, is provided in Section 18 (AI Decision Points).

### 6.3 Improved Customer Journey

A prospective business is researched and contacted automatically, follows up if unanswered, and — once onboarded — gains a self-service portal covering request submission and history, task status, website health, billing, and an instant AI-answered question channel. The client's entire relationship with Hamish AI is visible in one place, without needing to contact the operator for a status update.

### 6.4 Operational Improvements and Expected Efficiencies

- Elimination of manual first-pass triage for every incoming request.
- Automatic task and calendar creation removes a manual scheduling step for every piece of real work.
- Automatic invoice status reconciliation removes manual payment-confirmation chasing.
- One-click overdue reminders replace ad hoc, easily-forgotten chase emails.
- Weekly automated lead research removes a recurring manual prospecting task.
- A single Overview dashboard replaces the need to check four separate areas of the console for outstanding work.

Full efficiency and benefit quantification: Section 26 (Process Metrics) and Section 27 (Benefits Realisation).

## 7. Gap Analysis

Comparison of current (Section 5) and future (Section 6) state across five dimensions, with any residual gap explicitly called out rather than implied.

| Dimension | Current-State Gap                        | Future-State Closure                                                                        | Residual Gap                                                               |
|-----------|------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Process   | No formal request lifecycle or ownership | Fully modelled request lifecycle (new / awaiting_info / triaged) with conditional auto-send | No formal SLA enforcement or escalation beyond the Overview dashboard flag |

| Dimension  | Current-State Gap                                                           | Future-State Closure                                                                                 | Residual Gap                                                                                                    |
|------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Technology | No central system of record                                                 | Single platform (admin console and client portal) backed by one data store                           | Single-operator-only authentication model; no multi-user roles yet                                              |
| Data       | No structured lead, request or invoice data                                 | Structured relational schema across eight core entities (Section 15)                                 | No historical back-fill of pre-platform activity; data starts at go-live (except the migrated lead spreadsheet) |
| Capability | Manual, ad hoc judgement calls standing in for triage, drafting and scoring | Five AI decision points (Section 18) automate these judgement calls                                  | Prompt tuning will be required as new request categories emerge over time                                       |
| Skills     | No dedicated finance/ops background; reliance on operator memory            | Conservative auto-send thresholds and human-in-the-loop defaults reduce reliance on specialist skill | Operator remains the single point of expertise on how the system operates (tracked as RISK-02)                  |

## 8. Business Requirements

Each requirement is traceable to Section 30 (Traceability Matrix) via its ID and, where applicable, to the process numbering used in the Business Process Model (P1–P7).

| ID    | Description                                                                                          | Justification                                         | Pri.   | Source             | Acceptance Criteria                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-01 | Capture and centrally record every client request regardless of channel (portal or monitored inbox). | Eliminates lost or forgotten requests (Section 5.2).  | Must   | As-Is pain point   | A request submitted via the portal or received via the monitored inbox appears in the request log with raw text, timestamp and client link.                        |
| BR-02 | Automatically triage incoming requests to determine category, complexity and priority.               | Removes manual first-pass judgement; speeds response. | Must   | Objective 2.1      | Every new request is assigned a category, complexity and priority without manual entry.                                                                            |
| BR-03 | Automatically respond to requests that are unambiguous and covered by the client's maintenance plan. | Reduces operator time on routine work.                | Should | Objective 2.1      | Requests meeting the confidence-and-coverage rule (RULE-01) are marked auto-sent with a client-facing reply, without operator action.                              |
| BR-04 | Identify and surface requests that lack sufficient information before work can begin.                | Avoids wasted effort and back-and-forth delay.        | Must   | As-Is inefficiency | Requests classified as awaiting information show explicit clarifying questions to the client.                                                                      |
| BR-05 | Convert genuine work items into trackable tasks with a target completion date.                       | Prevents dropped commitments (Section 5.2).           | Must   | As-Is pain point   | A task is created for any request that cannot be auto-resolved but is sufficiently specified, linked to its source request, with a due date derived from priority. |

| ID    | Description                                                                                                                | Justification                                        | Pri.   | Source           | Acceptance Criteria                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|--------|------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| BR-06 | Notify clients automatically when their task is completed.                                                                 | Maintains client visibility without manual effort.   | Should | Objective 2.2    | A task status change to done triggers a client email.                                                                                    |
| BR-07 | Research and record prospective new-business leads on a recurring schedule.                                                | Removes a recurring manual prospecting task.         | Must   | Objective 2.1    | A scheduled process adds new prospect records (business name, category, location, contact email where available) at least weekly.        |
| BR-08 | Generate personalised outreach communications for qualified leads.                                                         | Keeps outreach fast without losing a personal touch. | Should | Objective 2.1    | An outreach email draft is produced referencing a relevant past case study where one applies.                                            |
| BR-09 | Flag leads that have not responded within a defined period for follow-up.                                                  | Prevents leads going cold (Section 5.2).             | Must   | As-Is pain point | A lead in contacted status with no status change after 5 days is flagged needs follow-up.                                                |
| BR-10 | Raise and deliver invoices to clients electronically with a payable link.                                                  | Removes manual, ad hoc invoicing.                    | Must   | As-Is pain point | An invoice created in the admin console results in a real payment-processor invoice and a client email containing a hosted payment link. |
| BR-11 | Automatically reconcile invoice payment status.                                                                            | Removes manual payment-confirmation chasing.         | Must   | As-Is pain point | A webhook event from the payment processor updates the invoice's stored status without manual intervention.                              |
| BR-12 | Automatically prompt clients to pay overdue invoices.                                                                      | Improves cash-flow predictability.                   | Should | Objective 2.1    | An invoice past its due date and still unpaid can be flagged, and a reminder sent with one action, with the send time recorded.          |
| BR-13 | Monitor client websites for availability and integrity issues on a recurring basis.                                        | Detects issues before the client does (Section 5.2). | Should | As-Is pain point | A scheduled process records uptime, SSL validity and broken-link status per client website at least daily.                               |
| BR-14 | Alert the operator when a monitored website issue is detected.                                                             | Enables proactive client communication.              | Must   | As-Is pain point | An unhealthy check result triggers an email alert to the operator.                                                                       |
| BR-15 | Provide clients with self-service visibility of their own request history, task status, website health and billing status. | Reduces unnecessary contact volume (Section 5.2).    | Must   | As-Is pain point | An authenticated client can view their own data in the portal, and only their own data.                                                  |
| BR-16 | Provide clients with an instant, AI-answered channel for common questions.                                                 | Reduces response latency for simple queries.         | Could  | Objective 2.2    | A client question submitted in the portal returns a drafted answer from a curated knowledge base without operator involvement.           |

| ID    | Description                                                                                          | Justification                                               | Pri.   | Source                 | Acceptance Criteria                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| BR-17 | Consolidate all outstanding operational items requiring operator attention into a single daily view. | Reduces context-switching across the console (Section 6.4). | Should | Objective 2.1          | Overdue invoices, stuck requests, stale leads and site issues are all visible from one landing page.                                 |
| BR-18 | Document the end-to-end operating model as a maintained process model and documentation pack.        | Enables governance, onboarding and audit.                   | Should | Governance requirement | This document and the associated Business Process Model exist, are internally consistent, and are kept aligned with the live system. |
| BR-19 | Allow the operator to access the internal console without depending solely on a shared password.     | Reduces daily access friction.                              | Could  | Objective 2.1          | An authorised operator can request a one-time sign-in link to a pre-approved address and be signed in without typing the password.   |

## 9. Functional Requirements

Each functional requirement implements one or more business requirements above. Inputs, outputs, business rules and exception handling are stated per function; detailed validations are covered where they materially affect behaviour.

| ID    | Function                                                            | Inputs                                                                                     | Outputs                                                                                                  | Business Rules / Exceptions                                                                                                                                                 |
|-------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | Log and triage an incoming request (implements BR-01, BR-02)        | Raw request text; source channel (portal form or monitored inbox message); client identity | Request record with category, complexity, priority and maintenance-coverage determination                | A monitored-inbox message is matched to a client by sender email; unmatched senders are not triaged. A message already labelled as processed is never re-triaged (RULE-10). |
| FR-02 | Determine auto-send or clarification flag (implements BR-03, BR-04) | Triage output (category, complexity, coverage determination)                               | Either an auto-sent client reply, or an awaiting_info status with explicit missing-information questions | Auto-send requires high confidence AND maintenance coverage (RULE-01); any other outcome defaults to awaiting_info or task creation, never a silent drop.                   |
| FR-03 | Create a task from a triaged request (implements BR-05)             | Triaged request classified as requiring real work                                          | Task record linked to the source request, with a title, description and due date                         | Due date is derived from priority per RULE-03. A request may generate at most one active task.                                                                              |
| FR-04 | Synchronise a task to the operator's calendar (implements BR-05)    | New task record                                                                            | A calendar event with a due date, title referencing the client, and a link back to the request           | If calendar sync is not configured (no valid credential), the task is still created; only the calendar step is skipped.                                                     |
| FR-05 | Notify the client on task completion (implements BR-06)             | Task status change to done                                                                 | Client email confirming completion                                                                       | Only a transition to done triggers an email; todo and in_progress transitions are silent, as the client has no action to take on them.                                      |

| ID    | Function                                                                    | Inputs                                                                    | Outputs                                                                                                                      | Business Rules / Exceptions                                                                                                                                                                                       |
|-------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-06 | Run scheduled lead research (implements BR-07)                              | Defined geographic scope (Central Belt of Scotland) and category criteria | New prospect records: business name, category, neighbourhood, website, contact email (where found), initial score and signal | Duplicate businesses already present are not re-added; a new prospect defaults to needs_verification status.                                                                                                      |
| FR-07 | Draft a personalised outreach email (implements BR-08)                      | Prospect record (category, business name, signal)                         | Draft subject and body, opened ready to send in the operator's email client                                                  | The prospect's category is matched against a maintained case-study list (Section 18, Decision 4); no match is forced if none is a confident fit.                                                                  |
| FR-08 | Flag a stale contacted lead for follow-up (implements BR-09)                | Prospect status and last-contacted timestamp                              | A visible needs-follow-up flag on the lead                                                                                   | Applies only where status = contacted and more than 5 days have elapsed since contacted_at (RULE-04).                                                                                                             |
| FR-09 | Create and send a client invoice (implements BR-10)                         | Client selection, amount, description (entered by operator)               | A payment-processor invoice, a stored invoice record, and a client email containing the hosted payment link                  | An invoice cannot be created for a client without a verified email on file (RULE-07). Due date defaults to 14 days from creation (RULE-06).                                                                       |
| FR-10 | Reconcile invoice status via payment webhook (implements BR-11)             | Signed webhook event from the payment processor                           | Updated invoice status (paid / void / uncollectible / payment failed) and, where paid, a paid timestamp                      | The webhook signature must verify before any data is trusted or written; unverifiable events are rejected outright.                                                                                               |
| FR-11 | Send an overdue-invoice reminder (implements BR-12)                         | Operator-initiated action on an invoice flagged overdue                   | A reminder email to the client with the payment link, and a recorded reminder-sent timestamp                                 | Only invoices with status = open and a due date in the past may be reminded (RULE-05); the action is one-click but never automatic, so the operator can first confirm the client hasn't already paid out-of-band. |
| FR-12 | Run a daily site-health check and alert on issues (implements BR-13, BR-14) | Each client's registered website URL                                      | A stored check result (uptime, SSL validity, broken links, AI summary) and, if unhealthy, an alert email                     | Only clients with a website URL on file are checked; a healthy result updates the portal summary silently.                                                                                                        |
| FR-13 | Present a client's own data in the self-service portal (implements BR-15)   | Authenticated client session                                              | Request history, task status, website-health summary and invoice/billing status, scoped to that client only                  | Only fields appropriate for client visibility are ever returned; internal-only fields (see Section 14, RULE-08) are excluded at the query level, not merely hidden in the interface.                              |
| FR-14 | Answer a client question from the knowledge base (implements BR-16)         | Client's typed question                                                   | A drafted answer sourced only from matched knowledge-base entries                                                            | If no relevant knowledge-base entry exists, the answer indicates this rather than fabricating a response.                                                                                                         |

| ID    | Function                                                       | Inputs                                                                        | Outputs                                                                                                                     | Business Rules / Exceptions                                                                                                                                                                                                    |
|-------|----------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-15 | Present the consolidated Overview dashboard (implements BR-17) | Current state of invoices, requests, leads and site checks across all clients | Counts and itemised lists of overdue invoices, stuck requests, stale leads and site issues, or confirmation that none exist | Each list links directly to the underlying record so the operator can act immediately.                                                                                                                                         |
| FR-16 | Provide passwordless admin sign-in (implements BR-19)          | Operator-submitted email address                                              | A one-time sign-in link emailed to that address; on click, an authenticated admin session                                   | The link only grants access if the submitted address exactly matches the single pre-approved administrator address (RULE-09); any other address is rejected silently, without confirming or denying whether it was recognised. |

## 10. Non-Functional Requirements

| ID     | Category     | Requirement                                                                                                                            | Rationale / Measure                                                                                 |
|--------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| NFR-01 | Performance  | AI triage shall return a result within a few seconds of request submission.                                                            | A fast, low-latency model is used specifically so the portal confirmation is not perceived as hung. |
| NFR-02 | Performance  | Scheduled batch jobs (inbox ingestion, site checks, digests) shall complete within the hosting platform's serverless execution window. | Prevents partial or failed daily runs.                                                              |
| NFR-03 | Security     | All client and business data shall be stored with Row Level Security enabled and zero public-read policies.                            | Access is only ever via a server-side credential, never exposed to the browser.                     |
| NFR-04 | Security     | Client portal authentication shall use passwordless, single-use magic links tied to a verified email address.                          | No long-lived shared client credential exists to be compromised.                                    |
| NFR-05 | Security     | Administrative access shall be gated by a shared secret or a magic link restricted to one pre-approved address.                        | No open self-registration exists for the admin console (RULE-09).                                   |
| NFR-06 | Security     | Inbound payment webhook calls shall be cryptographically verified against a shared signing secret before any data is trusted.          | Prevents forged payment-status events.                                                              |
| NFR-07 | Availability | Scheduled automations shall be resilient to the hosting platform's plan-tier scheduling constraints by design.                         | The platform assumes daily-only execution rather than depending on higher frequency.                |
| NFR-08 | Scalability  | The data model shall support a growing number of clients, requests and prospects without structural change.                            | Normalised relational schema with indexed foreign-key relationships (Section 15).                   |

| ID     | Category        | Requirement                                                                                                                                       | Rationale / Measure                                                            |
|--------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| NFR-09 | Accessibility   | Interfaces shall use semantic, keyboard-navigable controls; status shall never be conveyed by colour alone.                                       | Status is paired with an icon and text label in every case.                    |
| NFR-10 | Reliability     | Failure of a non-critical downstream integration (e.g. AI drafting) shall degrade gracefully.                                                     | A request can still be logged and reviewed manually even if AI drafting fails. |
| NFR-11 | Maintainability | Shared business logic (e.g. invoice-overdue and lead-follow-up determination) shall be implemented once and reused everywhere it is displayed.    | Prevents divergent behaviour between the admin console and the client portal.  |
| NFR-12 | Auditability    | Every invoice, request and processed email shall retain a timestamped lifecycle record sufficient to reconstruct history.                         | Supports the Issues Log (Section 22) and general governance.                   |
| NFR-13 | Compliance      | Client communications shall be sent from a domain-verified sender address and retain a clear human contact path even where content is AI-drafted. | Preserves trust and deliverability (see ISS-03).                               |
| NFR-14 | Compliance      | Payment processing shall rely on a PCI-compliant third-party processor; the platform shall never store or transmit raw card data.                 | Card data never touches Hamish AI's own systems.                               |

## 11. User Stories

| ID    | Story                                                                                                                   | Acceptance Criteria                                                     | Value  | Pri.   | Dependencies |
|-------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------|--------|--------------|
| US-01 | As a client, I want to submit a request through my portal so that I don't need to email or call.                        | Submitted text appears in my request history immediately with a status. | High   | Must   | BR-01, FR-01 |
| US-02 | As Hamish, I want incoming requests automatically categorised and prioritised so I don't triage every message by hand.  | Category, complexity and priority are populated without manual entry.   | High   | Must   | FR-01        |
| US-03 | As a client, I want simple, maintenance-covered requests handled automatically so I get a fast response.                | I receive a reply without visible delay when my request qualifies.      | High   | Should | FR-02        |
| US-04 | As a client, I want to be told exactly what information is missing if my request can't proceed, so I know what to send. | Specific clarifying questions are shown against my request.             | Medium | Must   | FR-02        |

| ID    | Story                                                                                                                 | Acceptance Criteria                                                                                                            | Value   | Pri.   | Dependencies |
|-------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------|--------|--------------|
| US-05 | As Hamish, I want genuine work automatically turned into a task with a calendar reminder so nothing slips through.    | A task and calendar event appear the moment triage classifies a request as real work.                                          | High    | Must   | FR-03, FR-04 |
| US-06 | As a client, I want to be told when my task is finished.                                                              | I receive an email the moment the task is marked done.                                                                         | Medium  | Should | FR-05        |
| US-07 | As Hamish, I want a weekly list of new prospective clients researched for me.                                         | New prospect records appear weekly without manual searching.                                                                   | High    | Must   | FR-06        |
| US-08 | As Hamish, I want a ready-to-send, personalised outreach email drafted for each qualified lead.                       | Clicking Email produces a subject/body referencing a relevant case study, opened ready to send.                                | High    | Should | FR-07        |
| US-09 | As Hamish, I want to be reminded to follow up with leads who haven't replied.                                         | A lead is visibly flagged needs follow-up after 5 days of silence.                                                             | Medium  | Must   | FR-08        |
| US-10 | As Hamish, I want to raise a client invoice from the same screen I manage their account.                              | Entering an amount and description on the client's page produces a real payable invoice.                                       | High    | Must   | FR-09        |
| US-11 | As Hamish, I want invoice status to update itself when a client pays.                                                 | Invoice status flips to paid without me touching it, upon payment.                                                             | High    | Must   | FR-10        |
| US-12 | As Hamish, I want a one-click way to nudge a client about an overdue invoice.                                         | An Overdue badge and Send reminder button appear automatically; clicking sends the email and timestamps it.                    | Medium  | Should | FR-11        |
| US-13 | As Hamish, I want to know if a client's website goes down or has an SSL/broken-link problem before they tell me.      | I receive an alert email the same day an issue is detected.                                                                    | High    | Should | FR-12        |
| US-14 | As a client, I want to see the status of everything I've asked for, my invoices, and my website health in one place.  | My portal shows request history, task status, invoices, and a website-health summary, and nothing belonging to another client. | High    | Must   | FR-13        |
| US-15 | As a client, I want to ask a quick question and get an instant answer.                                                | A typed question returns a drafted answer within seconds.                                                                      | Medium  | Could  | FR-14        |
| US-16 | As Hamish, I want one screen showing everything needing my attention across billing, requests, leads and site health. | The Overview page lists overdue invoices, stuck requests, stale leads and site issues, or confirms nothing is outstanding.     | High    | Should | FR-15        |
| US-17 | As Hamish, I want to get into the admin console without typing a password every time.                                 | I can request an emailed link to a pre-approved address and be signed in on click.                                             | Low-Med | Could  | FR-16        |

## 12. Use Cases

### UC-01 — Submit and Triage a Client Request

**Actors:** Client; AI Triage Engine; Business Owner (Operator)

**Trigger:** A client submits a request through the portal, or emails Hamish AI directly.

**Preconditions:** The client has an active portal account, or is a recognised sender in the monitored inbox.

#### Main Flow

- 1. Client enters free-text describing what they need (portal form) or sends an email.
- 2. If received by email, the daily inbox job matches the sender to a client record and imports the message as a request.
- 3. The AI Triage Engine reads the raw text and the client's maintenance-plan coverage.
- 4. The engine assigns a category, complexity and priority, and determines whether the request is covered by maintenance.
- 5. The engine determines whether the request is auto-sendable, needs clarification, or requires a task (see UC-02).
- 6. The request record is saved with its full triage output for later reference.

#### Alternative Flows

- Portal submission: the request is available for triage immediately, without waiting for the next scheduled inbox run.

#### Exception Flows

- If the AI Triage Engine is unavailable, the request is still logged with the raw text, and the operator is left to triage it manually (NFR-10).
- If an inbound email cannot be matched to any client, it is not imported as a request.

*Post-conditions: A request record exists with a definite status (new, awaiting\_info, or triaged) and, where applicable, a linked task.*

### UC-02 — Auto-Resolve or Flag a Request for Clarification

**Actors:** AI Triage Engine; Client; Business Owner (Operator)

**Trigger:** A request has just been triaged (continues from UC-01, step 5).

**Preconditions:** Triage output (category, complexity, priority, maintenance coverage) is available for the request.

#### Main Flow

- 1. The engine evaluates whether the request meets the auto-send rule: high confidence AND covered by maintenance (RULE-01).
- 2. If the rule is met, a client-facing reply is drafted and sent automatically; the request is marked auto\_sent with a response timestamp.
- 3. If the request is missing information needed to act, it is set to awaiting\_info and specific clarifying questions are shown to the client.

- 4. If neither applies, the request proceeds to task creation (UC not separately numbered — see Functional Requirement FR-03).

### Alternative Flows

- The client responds to an awaiting\_info request with the missing detail; the request re-enters triage (effectively restarting UC-01 from step 3).

### Exception Flows

- If the auto-send email fails to deliver, the failure is logged and the request is not marked as responded to, so it remains visible for manual follow-up.

*Post-conditions: The client has received either an automatic reply, a set of clarifying questions, or (via FR-03) a task has been created; in no case is the request left with no visible outcome.*

## UC-03 — Raise and Collect Payment for an Invoice

**Actors: Business Owner (Operator); Client; Payment Processor**

Trigger: The operator decides to bill a client for completed or agreed work.

Preconditions: The client has a verified email address on file (RULE-07).

### Main Flow

1. The operator enters an amount and description against the client's record.
2. The platform creates (or reuses) a payment-processor customer record for the client.
3. The platform creates and finalises an invoice with the payment processor, due in 14 days (RULE-06).
4. The platform stores an invoice record locally, linked to the client (and, where applicable, the originating request).
5. The platform emails the client a payment link, in the operator's own voice rather than the processor's default template.
6. The client opens the link and pays via the processor's hosted payment page.
7. The payment processor sends a signed webhook event confirming payment; the platform verifies the signature and marks the invoice paid, recording the payment timestamp.

### Alternative Flows

- The client does not pay before the due date: the invoice is treated as overdue (RULE-05) and appears on the Overview dashboard and the client detail page for a one-click reminder (UC not separately numbered — see FR-11).

### Exception Flows

- If invoice creation with the payment processor fails, no local invoice record is created and the operator is shown an explicit error rather than a false success.
- If the webhook signature does not verify, the event is rejected and no invoice status is changed.

*Post-conditions: A real, payable invoice exists with the payment processor and locally, its status is accurately reflected, and the client has been notified with a working payment link.*

## UC-04 — Research and Convert a Prospect to a Client

**Actors:** Scheduled Research Process; Business Owner (Operator); Prospect

**Trigger:** The weekly scheduled lead-research process runs.

**Preconditions:** None (this is the entry point of the acquisition pipeline).

### Main Flow

- 1. The scheduled process researches businesses in the Central Belt of Scotland matching defined categories.
- 2. Candidate businesses are recorded as prospects with a status of needs\_verification.
- 3. The operator reviews and verifies candidates, moving qualified ones to ready.
- 4. The operator requests an AI-drafted outreach email for a ready prospect; the engine matches the prospect's category to a relevant case study where one exists.
- 5. The operator reviews and sends the drafted email, and marks the prospect contacted, which records the contact timestamp.
- 6. If the prospect responds positively, the operator adds them as a client (Process 2 — Client Onboarding), completing the conversion.

### Alternative Flows

- No case-study match exists for the prospect's category: the outreach draft omits a proof point rather than forcing an irrelevant comparison.

### Exception Flows

- The prospect does not respond within 5 days: the lead is flagged needs follow-up (RULE-04) and reappears on the Overview dashboard until re-contacted or marked not a fit.

*Post-conditions: The prospect is either converted to a client, marked not a fit, or remains visibly tracked for follow-up — no lead is silently forgotten.*

## UC-05 — Monitor a Client Website and Alert on an Issue

**Actors:** Scheduled Monitoring Process; Business Owner (Operator); Client (indirect)

**Trigger:** The daily scheduled site-check process runs.

**Preconditions:** The client has a website URL on file.

### Main Flow

- 1. The scheduled process checks each client website's uptime, SSL certificate validity, and for broken links.
- 2. The raw results are stored as a site-check record with a timestamp.
- 3. If any check is unhealthy, an alert email is sent to the operator the same day.
- 4. If all checks are healthy, an AI-generated plain-English summary is stored for display in the client portal.

### Alternative Flows

- A client without a website URL on file is skipped entirely — this use case does not apply to them.

## Exception Flows

- If the target website itself is unreachable in a way that also prevents the check from completing, the failure is recorded rather than assumed healthy.

*Post-conditions: The operator is aware of any website issue before the client typically would be, and the client portal reflects an accurate, current health summary.*

## UC-06 — Client Self-Service Portal Session

**Actors: Client; Supabase Auth (identity provider)**

Trigger: A client requests a sign-in link for the portal.

Preconditions: The client's email exactly matches the email on their client record (RULE-08 governs data scoping once signed in).

### Main Flow

- 1. The client enters their email on the portal login page and requests a link.
- 2. A one-time magic link is emailed to that address.
- 3. The client clicks the link; the platform exchanges it for an authenticated session.
- 4. The platform matches the authenticated email to a client record.
- 5. The client views their own request history, task status, website-health summary and billing status.
- 6. The client may submit a new request or ask an instant question (UC-01 and FR-14 respectively) from within the same session.

### Alternative Flows

- The client asks an instant question via the knowledge base rather than submitting a full request.

## Exception Flows

- The authenticated email does not match any client record: the client sees an explicit no-access message rather than an error or another client's data.

*Post-conditions: The client has a working session scoped strictly to their own data, per RULE-08.*

# 13. Process Narratives

The following narratives walk through each of the seven processes exactly as modelled in the HamishAI Business Process Model, referencing the same node labels used in the diagram so the two artefacts can be read side by side.

## 13.1 Process 1 — Lead Generation

A weekly scheduled research task canvasses the Central Belt of Scotland for candidate businesses and writes them into the prospects table. Each prospect is checked for whether it has been verified and scored; unverified prospects return to the table for review, while verified, ready prospects proceed to an AI-drafted outreach email, personalised with a category-matched case study where one exists. The operator sends this email via Gmail. If the prospect has not replied within 5 days, the lead is flagged as needing follow-up and a further outreach draft is prepared; otherwise, once the prospect signs on, the flow continues into Process 2 (Client Onboarding).

## 13.2 Process 2 — Client Onboarding

The operator adds a client record, which immediately grants portal access — matched by email address, with no separate sign-up step required — and sets the client's maintenance plan (none, basic, or growth). From here, the client can submit requests (Process 3) and ask support questions (Process 4), and the maintenance plan determines how the client is billed (Process 6).

## 13.3 Process 3 — Request Handling & Fulfillment

A client submits a request via the portal, or emails Hamish AI directly, in which case the daily inbox cron picks it up. Either way, the request reaches AI triage, which determines category, complexity and priority. Triage then asks whether the request is both high-confidence and covered by the client's maintenance plan: if so, a reply is auto-sent to the client; if information is missing, the request is set to `awaiting_info` with explicit questions shown to the client, who can reply to re-enter the flow; if the request needs real work, a task is created, a calendar event is added with a due date set by priority, the operator completes the work, and the client is emailed that it is done.

## 13.4 Process 4 — Client Support

A client asks a question directly in the portal. The platform looks the question up against the curated knowledge base and returns an AI-drafted instant answer, without requiring operator involvement.

## 13.5 Process 5 — Site Monitoring

A daily site-check cron examines each client's website. If an issue is found — the site is down, its SSL certificate is invalid, or it has broken links — an alert is emailed to the operator. If the site is healthy, a plain-English summary is generated and shown in the client's portal.

## 13.6 Process 6 — Billing

The operator creates an invoice, which is created and finalised with the payment processor and emailed to the client as a payment link. The platform then asks whether the invoice is paid before its due date: if so, a webhook from the payment processor marks it paid, and its status becomes visible in the client's portal billing page; if the invoice becomes overdue, a reminder email is sent, returning the flow to the payment-link step until the invoice is settled.

## 13.7 Process 7 — Daily Overview

Four streams feed a single consolidated view for the operator: overdue invoices, requests stuck on `awaiting_info`, leads needing follow-up, and site issues. Each of these is itself the output of a decision point elsewhere in the model — the `awaiting_info` branch of Process 3, the overdue branch of Process 6, the follow-up flag of Process 1, and the issue-found branch of Process 5 — and each flows, dotted in the diagram to indicate a cross-cutting signal rather than a sequential step, into the `/admin` Overview dashboard.

## 14. Business Rules

| ID      | Rule                                                                                                                                                                                                                                                            |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RULE-01 | A request is only auto-sent if triage confidence is high AND the request is covered by the client's maintenance plan.                                                                                                                                           |
| RULE-02 | A request lacking necessary information is set to awaiting_info and must show explicit missing-information questions to the client.                                                                                                                             |
| RULE-03 | Task due dates are derived from priority: urgent = +1 day, high = +2 days, medium = +5 days, low = +10 days from creation.                                                                                                                                      |
| RULE-04 | A lead is flagged for follow-up only when its status is contacted AND more than 5 days have elapsed since it was last contacted.                                                                                                                                |
| RULE-05 | An invoice is considered overdue only when its status is open AND its due date has passed; paid, void or uncollectible invoices are never shown as overdue.                                                                                                     |
| RULE-06 | An invoice's due date defaults to 14 days from creation.                                                                                                                                                                                                        |
| RULE-07 | An invoice cannot be created for a client without a verified email address on file.                                                                                                                                                                             |
| RULE-08 | A client's portal session grants visibility only into records where the client identifier matches the authenticated client's own record; data belonging to any other client is never returned.                                                                  |
| RULE-09 | Only the single pre-approved administrator email address may complete an admin magic-link sign-in; a session under any other address is discarded without granting access.                                                                                      |
| RULE-10 | A monitored inbox message is processed at most once — a processed message is labelled and excluded from all future scans, regardless of triage outcome.                                                                                                         |
| RULE-11 | A client-facing email is only ever sent from the domain-verified sender address, never a personal inbox.                                                                                                                                                        |
| RULE-12 | Overdue-invoice reminder emails use a fixed, non-AI-generated template; prospect outreach emails use an AI-personalised draft — the two are deliberately different in generation method given their different purpose (factual notice versus persuasive pitch). |

## 15. Data Requirements

| Entity               | Key Attributes                                                                                                              | Source                                            | Validation                                                      | Ownership                         | Retention                                                   | Quality Requirement                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Client               | business_name, contact name, email, package, maintenance_plan, website_url, stripe_customer_id                              | Operator entry (onboarding)                       | Email format; uniqueness against portal-matching rule (RULE-08) | Business Owner                    | Retained for the life of the client relationship            | Email must exactly match the address the client will sign in with                       |
| Request              | raw_text, status, category, complexity, priority, missing_info, coverage_reasoning, draft_response, auto_sent, responded_at | Client (portal or inbox) / AI Triage Engine       | Must be linked to a valid client                                | Business Owner                    | Retained indefinitely as operational history                | Every request has a definite status at all times; never left unclassified               |
| Task                 | title, description, acceptance_criteria, status, calendar_event_id, request_id                                              | AI Triage Engine (creation) / Operator (progress) | Must be linked to a valid request                               | Business Owner                    | Retained indefinitely as operational history                | Status is one of todo / in_progress / done at all times                                 |
| Prospect             | business_name, category, neighbourhood, website, email, score, signal, outreach_note, status, contacted_at, found_at        | Scheduled research process / Operator             | Score constrained to 0–5 where set                              | Business Owner                    | Retained while the prospect is active or recently contacted | Status is one of needs_verification / ready / contacted / not_fit                       |
| Invoice              | amount_pence, description, status, due_date, paid_at, reminder_sent_at, stripe_invoice_id, stripe_hosted_invoice_url        | Operator entry / Payment processor webhook        | Client must have a verified email (RULE-07)                     | Business Owner (financial record) | Retained indefinitely for financial audit                   | Status accurately reflects the payment processor's own record at all times              |
| Site Check           | checked_at, uptime_ok, ssl_ok, ssl_valid_until, response_ms, broken_links, ai_summary                                       | Scheduled monitoring process                      | Must be linked to a client with a website URL                   | Business Owner                    | Recent history retained for trend visibility                | Only the latest check per client is surfaced as the current health status               |
| Knowledge Base Entry | title, content, client_id (nullable for general entries)                                                                    | Operator curation                                 | Content must be reviewed before publication                     | Business Owner                    | Retained until superseded or removed                        | Entries are the sole source for portal AI answers — no answer is fabricated beyond them |

| Entity          | Key Attributes                               | Source                            | Validation                                                 | Ownership                    | Retention                               | Quality Requirement                                    |
|-----------------|----------------------------------------------|-----------------------------------|------------------------------------------------------------|------------------------------|-----------------------------------------|--------------------------------------------------------|
| Processed Email | message_id, client_id, subject, processed_at | Monitored inbox ingestion process | message_id must be unique (prevents reprocessing, RULE-10) | Business Owner (audit trail) | Retained as an audit trail of ingestion | Every triaged inbound email has a corresponding record |

## 16. Data Flow

Data enters the platform through three channels: direct client input (portal request and question submission), operator input (client onboarding, invoice creation, task progress), and scheduled automation (lead research, inbox ingestion, site checks). Every channel writes to the single Supabase-backed system of record described in Section 15; there is no secondary data store or spreadsheet in the target state.

From storage, data flows outward in two directions. First, into AI decision points (Section 18), which read request text, client coverage, prospect category and knowledge-base content to produce triage outcomes, drafts and answers that are written back to the same records. Second, into reporting surfaces: the client portal reads a client-scoped subset of the data (RULE-08); the admin console reads the full record set per client; and the Overview dashboard (Process 7) reads narrow, cross-client slices — overdue invoices, awaiting\_info requests, stale leads, and unhealthy site checks — aggregating them into one view without duplicating the underlying data.

Outbound data leaves the platform as email (client replies, task-completion notices, invoice links, reminders, alerts, outreach) via the transactional email provider, and as payment instructions to the payment processor, which in turn reports back via webhook. No client data is transmitted to any party beyond what each integration in Section 17 requires to perform its specific function.

## 17. Integration Requirements

| System                     | Purpose                                                                                             | Authentication                                                                             | Error Handling                                                                                                          | Synchronisation                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Anthropic Claude (AI)      | Request triage, drafting, portal Q&A, lead-email drafting and scoring                               | Server-side API key                                                                        | Failure is caught and logged; the underlying request/task/lead record is still saved for manual handling (NFR-10)       | Synchronous, per request                                                |
| Supabase (Postgres + Auth) | System of record for all entities; identity provider for the client portal and the admin magic-link | Service-role key (server-side data access) / anon key plus session cookie (authentication) | Query errors are logged server-side; the user is shown a generic fallback rather than a raw error                       | Real-time (direct queries; no batch synchronisation)                    |
| Stripe (Payments)          | Invoice issuance, hosted payment page, payment status events                                        | Secret API key (server-side) plus a webhook signing secret                                 | Webhook signature verification rejects any unverifiable call outright (NFR-06); invoice-creation errors are surfaced to | Event-driven via webhook (paid / void / uncollectible / payment failed) |

| System                       | Purpose                                                     | Authentication                     | Error Handling                                                                                                                                               | Synchronisation                                                                           |
|------------------------------|-------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|                              |                                                             |                                    | the operator rather than silently swallowed                                                                                                                  |                                                                                           |
| Google Gmail & Calendar APIs | Inbound email ingestion; calendar sync for tasks            | OAuth2 with a stored refresh token | A missing or expired token disables the affected feature gracefully — the function reports a configuration error rather than throwing an unhandled exception | Daily batch (Gmail ingestion); synchronous (calendar event creation)                      |
| Resend (Email Delivery)      | Transactional email delivery to clients and internal alerts | API key                            | Delivery failure is logged; the platform does not silently mark a message as sent if delivery failed                                                         | Synchronous, per send                                                                     |
| Vercel Cron                  | Scheduling of daily batch automations                       | Shared cron secret header          | N/A (scheduling infrastructure)                                                                                                                              | Daily, fixed times — a platform-tier constraint (Section 3.4) rather than a design choice |

## 18. AI Decision Points

Five points in the model involve an AI-made determination that materially affects what a client sees or receives. Each is documented with its inputs, logic, outputs, confidence handling, and the human override that remains available in every case.

### Decision 1 — Request Triage Classification

| Aspect              | Detail                                                                                                                                                                     |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs              | Raw request text; client profile (maintenance plan, package)                                                                                                               |
| Logic               | The AI model classifies the request into a category, complexity and priority, and determines whether it is covered by the client's maintenance plan                        |
| Outputs             | category, complexity, priority, covered_by_maintenance, coverage_reasoning                                                                                                 |
| Confidence Handling | There is no numeric confidence score exposed; instead, the auto-send gate (Decision 2) treats classification as usable only when the model itself signals a clear-cut case |
| Human Override      | The draft_response field remains editable by the operator regardless of the auto-send outcome; every task remains fully operator-executed                                  |

### Decision 2 — Auto-Send Eligibility

| Aspect | Detail               |
|--------|----------------------|
| Inputs | Output of Decision 1 |

| Aspect              | Detail                                                                                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| Logic               | A reply is auto-sent only if confidence is high AND the request is covered by maintenance (RULE-01)                       |
| Outputs             | auto_sent flag; responded_at timestamp; the client-facing reply itself                                                    |
| Confidence Handling | Deliberately conservative — defaults to the awaiting_info or task-creation path whenever the criteria are not clearly met |
| Human Override      | None automatic by design; the operator can always review request history and the reply that was sent                      |

### Decision 3 — Missing-Information Detection

| Aspect              | Detail                                                                                                       |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| Inputs              | Raw request text                                                                                             |
| Logic               | The model identifies what information is required to act on the request but is absent from the text supplied |
| Outputs             | A list of specific missing_info questions shown to the client                                                |
| Confidence Handling | Errs towards asking a clarifying question rather than guessing at intent                                     |
| Human Override      | The operator can still triage the request manually via the same admin tool at any time                       |

### Decision 4 — Lead Category-to-Case-Study Matching

| Aspect              | Detail                                                                                                                                                                                                                  |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs              | The lead's free-text category                                                                                                                                                                                           |
| Logic               | A keyword-based match against a maintained list (e.g. restaurant/cafe -> The Gannet, trades -> Craigie & Sons, hotel/B&B -> Assembly Rooms Hotel, gym/fitness -> Forge Fitness, professional services -> Lomond & Grey) |
| Outputs             | A matched case-study reference embedded verbatim in the outreach draft, or no proof point if no confident match exists                                                                                                  |
| Confidence Handling | Deliberately conservative — no match is forced for categories such as salons or generic retailers where no relevant case study exists                                                                                   |
| Human Override      | The operator reviews every draft before it is sent; nothing is auto-sent to a prospect                                                                                                                                  |

### Decision 5 — Knowledge-Base Answer Drafting

| Aspect | Detail                                                                              |
|--------|-------------------------------------------------------------------------------------|
| Inputs | The client's question; curated knowledge-base entries (general and client-specific) |

| Aspect              | Detail                                                                                                                                          |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Logic               | The model answers only from matched knowledge-base content                                                                                      |
| Outputs             | An instant answer shown in the portal                                                                                                           |
| Confidence Handling | If no relevant entry exists, the response reflects that rather than fabricating an answer                                                       |
| Human Override      | Not required for correctness, since answers are informational rather than transactional; the knowledge base itself is entirely operator-curated |

## 19. Risk Assessment

| ID      | Risk                                                                                                                                                                                | Prob.            | Impact  | Mitigation                                                                                                                                            | Owner  |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| RISK-01 | AI misclassifies a request and it is auto-sent incorrectly, giving a client a wrong or unhelpful reply.                                                                             | Low-Med          | Medium  | Conservative auto-send gate (RULE-01: high confidence AND maintenance-covered only); every response remains visible and auditable in request history. | Hamish |
| RISK-02 | Single-operator dependency — if Hamish is unavailable, no one else can act on flagged tasks or requests.                                                                            | Medium           | High    | The Overview dashboard consolidates urgent items for rapid catch-up; this documentation pack enables onboarding of a second operator if required.     | Hamish |
| RISK-03 | The Google Cloud OAuth application remains in Testing publishing status, causing the refresh token to expire every 7 days and silently disabling inbox ingestion and calendar sync. | High (recurring) | Medium  | A quick reconnect flow is documented at /admin/google-setup; recommend progressing the app to Verified status (Section 32).                           | Hamish |
| RISK-04 | The payment processor remains in sandbox/test mode; a configuration slip could allow test invoices to be mistaken for real ones, or vice versa when promoted to live.               | Low              | High    | An explicit go-live checklist step to rotate to live keys and re-verify the webhook endpoint before onboarding paying clients (Section 32).           | Hamish |
| RISK-05 | The hosting platform's entry-tier cron limits (daily only) delay time-sensitive automations, e.g. up to 24 hours' latency on inbox ingestion.                                       | High (by design) | Low-Med | The portal's direct-submission channel bypasses email-ingestion latency entirely for time-sensitive requests.                                         | Hamish |
| RISK-06 | A third-party outage (data platform, payment processor, AI provider, email delivery, calendar/inbox provider) disrupts one or more processes.                                       | Low per provider | Medium  | Failures are logged and degrade gracefully; no single integration failure takes down the whole platform (NFR-10).                                     | Hamish |

| ID      | Risk                                                                                                                     | Prob.   | Impact | Mitigation                                                                                                                     | Owner  |
|---------|--------------------------------------------------------------------------------------------------------------------------|---------|--------|--------------------------------------------------------------------------------------------------------------------------------|--------|
| RISK-07 | A client's registered email changes or is mistyped, breaking the email-matched portal-access model.                      | Low     | Medium | The admin console shows the exact email a client must use to sign in, and it is directly editable by the operator.             | Hamish |
| RISK-08 | Sensitive internal reasoning fields (coverage_reasoning, draft_response, priority) are accidentally exposed to a client. | Low     | Medium | The client portal deliberately selects only client-safe fields at the query level, verified during build and covered by FR-13. | Hamish |
| RISK-09 | An overdue-invoice reminder is sent after a client has already paid by an out-of-band method, landing as tone-deaf.      | Low-Med | Low    | The reminder is a manual one-click action, not automatic, so the operator can check first.                                     | Hamish |

## 20. Assumptions Log

| ID     | Assumption                                                                                                                  | Status                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| ASM-01 | Hamish remains the sole system operator for the documented scope of this platform.                                          | Holds                            |
| ASM-02 | All clients are UK-registered small businesses invoiced in GBP.                                                             | Holds                            |
| ASM-03 | Clients are willing and able to use a web-based portal; no dedicated client-side app support is required.                   | Holds                            |
| ASM-04 | The payment processor account will be promoted from test/sandbox to live mode before any real client is charged.            | Open — pending go-live           |
| ASM-05 | The Google Cloud OAuth application will either remain manually reconnected every 7 days or be moved out of Testing status.  | Open — see RISK-03               |
| ASM-06 | Client, request and lead volumes remain within the practical review capacity of a single operator for the documented scope. | Holds; no formal ceiling defined |

## 21. Dependency Log

| ID     | Dependency                                                   | Required For                                         |
|--------|--------------------------------------------------------------|------------------------------------------------------|
| DEP-01 | Anthropic API availability and key validity                  | All five AI decision points (Section 18)             |
| DEP-02 | Supabase project availability with schema migrations applied | All data persistence; both authentication mechanisms |

| ID     | Dependency                                                                                     | Required For                                           |
|--------|------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| DEP-03 | Payment processor account configuration (API key, webhook secret, registered webhook endpoint) | Process 6 — Billing                                    |
| DEP-04 | Google Cloud OAuth client with a valid, unexpired refresh token                                | Inbox ingestion and calendar sync (Process 3)          |
| DEP-05 | Email delivery provider domain verification on hamishai.org                                    | All client-facing and internal email (RULE-11)         |
| DEP-06 | Hosting platform cron configuration and shared cron secret                                     | All scheduled automations (Processes 1, 3, 5, 7)       |
| DEP-07 | DNS / domain resolution for hamishai.org and its hosting alias                                 | Platform reachability; OAuth and webhook redirect URLs |

## 22. Issues Log

The following issues were identified and resolved during delivery. They are retained here as a governance record rather than removed, since several materially shaped the final design (see cross-references to the affected Business/Functional Requirement and Business Rule).

| ID     | Issue                                                                                                                                                                | Resolution                                                                                                                                    | Status                           |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| ISS-01 | The hosting platform's entry-tier plan rejected a 30-minute cron schedule for inbox ingestion.                                                                       | Changed to a fixed daily time (see Section 3.4, Constraints).                                                                                 | Closed                           |
| ISS-02 | A pre-existing, unrelated table with the same intended name caused new lead records to silently fail to insert.                                                      | The new lead-tracking table was renamed to prospects throughout the schema and code.                                                          | Closed                           |
| ISS-03 | Client-facing email sent from a personal address was liable to rejection or spam-flagging for external recipients.                                                   | All client-facing email is now sent from a domain-verified address (RULE-11, NFR-13).                                                         | Closed                           |
| ISS-04 | A browser pop-up used to open a drafted outreach email was silently blocked when triggered from an asynchronous callback rather than directly from the user's click. | A placeholder tab is opened synchronously on click and redirected once the draft is ready.                                                    | Closed                           |
| ISS-05 | The payment processor's sandbox environment rejected the platform's own built-in invoice-send call outright.                                                         | The platform now emails the payment link itself via its own transactional email rather than relying on the processor's built-in send (FR-09). | Closed                           |
| ISS-06 | An early version of the admin navigation's active-state logic would have highlighted the Overview link as active on every page once it was added at the root path.   | An exact-match rule was added for the root navigation link before release.                                                                    | Closed (caught prior to release) |

| ID     | Issue                                                                                                                                                                               | Resolution                                                                                                            | Status |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|
| ISS-07 | An update to a status-tracking field silently failed to persist because a required schema column did not yet exist, and the write path did not surface the failure.                 | The schema column was added; error handling was added so any future occurrence is logged rather than silent (NFR-12). | Closed |
| ISS-08 | A duplicate hosting project, connected to the same source repository via a separate integration, was found to be silently deploying a full mirror of the platform on every release. | The duplicate project was identified and removed.                                                                     | Closed |

## 23. RAID Log

A consolidated view of Risks, Assumptions, Issues and Dependencies. Full detail for each entry is in Sections 19–22; this table exists for at-a-glance governance review.

| Type       | ID      | Summary                                                      | Status                            |
|------------|---------|--------------------------------------------------------------|-----------------------------------|
| Risk       | RISK-01 | AI misclassification leading to an incorrect auto-sent reply | Open — mitigated                  |
| Risk       | RISK-02 | Single-operator dependency (bus factor)                      | Open — mitigated                  |
| Risk       | RISK-03 | Google OAuth Testing-status token expiry (7-day cycle)       | Open — recurring, mitigated       |
| Risk       | RISK-04 | Payment processor sandbox vs live-mode configuration slip    | Open — mitigated, pending go-live |
| Risk       | RISK-05 | Daily-only cron scheduling latency                           | Open — accepted, mitigated        |
| Risk       | RISK-06 | Third-party provider outage                                  | Open — mitigated                  |
| Risk       | RISK-07 | Client email mismatch breaking portal access                 | Open — mitigated                  |
| Risk       | RISK-08 | Accidental exposure of internal-only fields to a client      | Open — mitigated                  |
| Risk       | RISK-09 | Overdue reminder sent after out-of-band payment              | Open — mitigated                  |
| Assumption | ASM-01  | Hamish remains sole operator                                 | Holds                             |
| Assumption | ASM-02  | All clients are UK-based, GBP-invoiced                       | Holds                             |
| Assumption | ASM-03  | Clients accept a web-based portal                            | Holds                             |
| Assumption | ASM-04  | Payment processor to be promoted to live mode before go-live | Open — pending                    |
| Assumption | ASM-05  | Google OAuth app reconnected or verified                     | Open — pending                    |
| Assumption | ASM-06  | Volumes remain within single-operator capacity               | Holds                             |

| Type       | ID              | Summary                                          | Status |
|------------|-----------------|--------------------------------------------------|--------|
| Issue      | ISS-01 – ISS-08 | Eight delivery issues, all resolved (Section 22) | Closed |
| Dependency | DEP-01 – DEP-07 | Seven external dependencies (Section 21)         | Active |

## 24. RACI Matrix

Because Hamish AI is a single-operator business, the Business Owner column is Accountable throughout and frequently also Responsible; this is documented explicitly rather than left to be inferred, per Section 4's stakeholder note.

| Activity                          | Business Owner | Client | AI Engine | Payment Processor | Data Platform |
|-----------------------------------|----------------|--------|-----------|-------------------|---------------|
| Request triage                    | A              | I      | R         | -                 | -             |
| Auto-send decision                | A              | I      | R         | -                 | -             |
| Task completion                   | R / A          | I      | -         | -                 | -             |
| Lead research & outreach drafting | A              | -      | R         | -                 | -             |
| Lead follow-up decision           | R / A          | -      | -         | -                 | -             |
| Invoice creation                  | R / A          | I      | -         | C                 | -             |
| Payment reconciliation            | A              | I      | -         | R                 | C             |
| Overdue reminder                  | R / A          | I      | -         | -                 | -             |
| Site health check                 | A              | I      | C         | -                 | R             |
| Portal Q&A                        | A              | C      | R         | -                 | C             |
| Overview dashboard review         | R / A          | -      | -         | -                 | C             |

*R = Responsible, A = Accountable, C = Consulted, I = Informed.*

## 25. MoSCoW Prioritisation

| Priority | Items                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Must     | BR-01, BR-02, BR-04, BR-05, BR-07, BR-09, BR-10, BR-11, BR-14, BR-15 — the core request, task, lead-capture, billing and site-alert capabilities without which the platform does not meet its purpose. |

| Priority             | Items                                                                                                                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Should               | BR-03, BR-06, BR-08, BR-12, BR-13, BR-17, BR-18 — automation that materially improves efficiency and client experience but where a manual fallback remains acceptable in the interim.                                                                                                                                                 |
| Could                | BR-16, BR-19 — enhancements that improve convenience (instant Q&A, passwordless admin access) without being essential to core operation.                                                                                                                                                                                              |
| Won't (this release) | Recurring/subscription invoicing for maintenance-plan clients; automated card-on-file auto-charge; multi-operator/role-based admin access; native mobile applications; multi-currency billing; formally contracted SLA guarantees — all explicitly out of scope per Section 3.2, several flagged as Phase 2 candidates in Section 32. |

## 26. Process Metrics

| KPI                             | Definition                                                                                         | Target (Phase 1 baseline)                                     | Data Source                         |
|---------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------|
| Auto-resolution rate            | Requests marked auto-sent as a proportion of all requests                                          | ≥ 25% within the first 90 days of operation                   | requests.auto_sent                  |
| Average time-to-first-response  | Time from request creation to auto-send or first operator action                                   | < 4 business hours                                            | requests.created_at / responded_at  |
| Awaiting-info resolution time   | Time a request spends in awaiting_info before the client replies                                   | Tracked, no fixed target in Phase 1 (baseline-setting)        | requests.status history             |
| Task completion lead time       | Time from task creation to status = done                                                           | Within the due date derived by RULE-03 for ≥ 80% of tasks     | tasks.created_at / status           |
| Lead-to-client conversion rate  | Prospects reaching contacted that become clients, as a proportion of all contacted                 | Tracked, no fixed target in Phase 1 (baseline-setting)        | prospects / clients                 |
| Follow-up compliance            | Proportion of leads flagged needs-follow-up that are actioned within 7 days of the flag            | ≥ 90%                                                         | prospects.contacted_at              |
| Invoice on-time payment rate    | Invoices paid before their due date, as a proportion of all invoices issued                        | ≥ 80%                                                         | invoices.due_date / paid_at         |
| Overdue reminder effectiveness  | Reminded invoices paid within 7 days of the reminder, as a proportion of all reminders sent        | Tracked, no fixed target in Phase 1 (baseline-setting)        | invoices.reminder_sent_at / paid_at |
| Site-check issue detection rate | Issues caught by monitoring versus issues first reported by the client                             | Monitoring should detect ≥ 90% of issues before client report | site_checks + qualitative log       |
| Portal adoption rate            | Distinct clients signing into the portal at least once, as a proportion of all clients with access | ≥ 70% within 60 days of onboarding                            | Supabase Auth sessions vs. clients  |

| KPI                        | Definition                                                                           | Target (Phase 1 baseline)                                                       | Data Source              |
|----------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------|
| Operator daily-review time | Time spent reviewing the Overview dashboard versus estimated pre-platform admin time | Directional reduction versus Section 5 baseline (qualitative, timed comparison) | Qualitative operator log |

## 27. Benefits Realisation

### 27.1 Financial Benefits

- Reduced incidence of late payment through automatic invoice reconciliation and one-click overdue reminders.
- Reduced non-billable admin hours previously spent on manual triage, scheduling and payment chasing.
- Reduced risk of lost new-business opportunity through automated lead follow-up flagging.

### 27.2 Operational Benefits

- A single source of truth for every client, request, task, lead and invoice.
- Reduced context-switching via the consolidated Overview dashboard (Process 7).
- Consistent, auditable client communication regardless of the channel a client used.

### 27.3 Customer Benefits

- Faster responses, including same-day resolution of routine, maintenance-covered requests.
- Transparent, self-service visibility of request status, task progress, website health and billing.
- Instant answers to common questions without waiting for a reply.
- Earlier notice of website issues, ahead of the client discovering them independently.

### 27.4 Strategic Benefits

- Hamish AI's own operation becomes a live reference case for the AI-automation services it sells to clients.
- A scalable operating model supports growth in client volume without a proportional increase in operator time.
- A documented, governed operating model (this pack) enables future onboarding of additional operators or investment discussions.

## 28. Wireframe Recommendations

The following component-level UI recommendations were made during analysis and have been delivered in the current build.

| Screen                  | Purpose                                    | Key Components                                                                                                                                                          |
|-------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Admin Overview (/admin) | Single daily landing page for the operator | Stat tiles (counts); grouped attention lists (overdue invoices, awaiting-info requests, stale leads, site issues); empty-state confirmation when nothing is outstanding |

| Screen                                 | Purpose                                        | Key Components                                                                                                                                                         |
|----------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Client Detail<br>(/admin/clients/[id]) | Full operational view of a single client       | Request history list with status badges; task list with due-date flags; invoice list with status badges and one-click reminder; progress-report and site-check actions |
| Leads Board<br>(/admin/leads)          | Pipeline management for prospective clients    | Status filter chips; needs-follow-up filter; per-lead outreach email action; inline contact-email editing                                                              |
| Portal Home (/portal)                  | Client self-service landing page               | Website-health summary card; instant-question box; new-request form; request history with client-safe status labels                                                    |
| Portal Billing<br>(/portal/billing)    | Client-facing billing visibility               | Invoice list with status badges (awaiting payment / paid / overdue) and a pay-now / view-receipt link                                                                  |
| Process Map<br>(/admin/process)        | Visual reference model of the entire operation | Zoomable, pannable swimlane diagram grouped by process, matching Section 13's narratives node-for-node                                                                 |

## 29. Test Scenarios

| ID    | Scenario                                                                            | Type           | Expected Result                                                                                                       |
|-------|-------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------|
| TS-01 | Client submits a clear, maintenance-covered request via the portal                  | Happy path     | Request is triaged, auto-sent, and appears in the client's request history with an auto-sent indicator (FR-01, FR-02) |
| TS-02 | Client submits a request missing key detail                                         | Happy path     | Request is set to awaiting_info with specific clarifying questions shown to the client (FR-02, RULE-02)               |
| TS-03 | Client emails Hamish AI directly instead of using the portal                        | Alternate path | The next daily inbox run imports the message as a request against the correct client (FR-01)                          |
| TS-04 | An email arrives from a sender not matched to any client                            | Edge case      | The message is not imported as a request and no error is raised to any client                                         |
| TS-05 | A request is classified as requiring real work                                      | Happy path     | A task is created with a due date matching RULE-03 for its priority, and a calendar event is added (FR-03, FR-04)     |
| TS-06 | Calendar sync credential is missing or invalid when a task is created               | Error handling | The task is still created; only the calendar step is skipped without failing the whole operation (FR-04)              |
| TS-07 | Operator marks a task done                                                          | Happy path     | The client receives a completion email; no email is sent for todo or in_progress transitions (FR-05)                  |
| TS-08 | Weekly lead research runs and finds a new candidate business                        | Happy path     | A new prospect record is created with status needs_verification (FR-06)                                               |
| TS-09 | Operator requests an outreach draft for a lead with no matching case study category | Edge case      | The draft is produced without a forced or irrelevant proof point (Decision 4, Section 18)                             |

| ID    | Scenario                                                                                | Type           | Expected Result                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| TS-10 | A contacted lead has had no status change for more than 5 days                          | Happy path     | The lead is flagged needs follow-up and appears on the Overview dashboard (RULE-04, FR-08)                                              |
| TS-11 | Operator creates an invoice for a client with no email on file                          | Error handling | Invoice creation is rejected with an explicit message; no payment-processor invoice is created (RULE-07)                                |
| TS-12 | Client pays an invoice via the hosted payment link                                      | Happy path     | A verified webhook event marks the invoice paid with a timestamp, visible in both the admin console and the client portal (FR-10)       |
| TS-13 | A webhook event arrives with an invalid signature                                       | Error handling | The event is rejected outright and no invoice status is changed (NFR-06)                                                                |
| TS-14 | An invoice passes its due date unpaid                                                   | Alternate path | The invoice displays an Overdue badge and a Send reminder action becomes available (RULE-05, FR-11)                                     |
| TS-15 | Operator sends an overdue reminder                                                      | Happy path     | The client receives a reminder email with the payment link, and reminder_sent_at is recorded (FR-11)                                    |
| TS-16 | Daily site check finds a client website is down                                         | Happy path     | An alert email is sent to the operator the same day (FR-12)                                                                             |
| TS-17 | Daily site check finds a client website healthy                                         | Happy path     | No alert is sent; an AI summary is stored for the client portal (FR-12)                                                                 |
| TS-18 | A client attempts to sign into the portal with an email that matches no client record   | Error handling | The client sees an explicit no-access message; no other client's data is exposed (RULE-08, UC-06)                                       |
| TS-19 | Client asks a question with no matching knowledge-base entry                            | Edge case      | The answer reflects that no relevant information was found, rather than fabricating a response (FR-14)                                  |
| TS-20 | An address other than the pre-approved administrator email requests an admin magic link | Error handling | No admin session is granted for that address, and the outcome is indistinguishable from a legitimate but unrecognised request (RULE-09) |

## 30. Traceability Matrix

Business Objective → Business Requirement → Functional Requirement → User Story → Test Scenario. A representative full-chain mapping is provided below; every Business Requirement in Section 8 traces through to at least one Functional Requirement, User Story and Test Scenario via this pattern.

| Business Objective             | Business Req. | Functional Req. | User Story   | Test Scenario       |
|--------------------------------|---------------|-----------------|--------------|---------------------|
| Scale without headcount growth | BR-01, BR-02  | FR-01           | US-01, US-02 | TS-01, TS-03, TS-04 |
| Scale without headcount growth | BR-03, BR-04  | FR-02           | US-03, US-04 | TS-01, TS-02        |

| Business Objective                   | Business Req. | Functional Req.     | User Story   | Test Scenario            |
|--------------------------------------|---------------|---------------------|--------------|--------------------------|
| Scale without headcount growth       | BR-05, BR-06  | FR-03, FR-04, FR-05 | US-05, US-06 | TS-05, TS-06, TS-07      |
| Repeatable acquisition pipeline      | BR-07, BR-08  | FR-06, FR-07        | US-07, US-08 | TS-08, TS-09             |
| Repeatable acquisition pipeline      | BR-09         | FR-08               | US-09        | TS-10                    |
| Cash-flow predictability             | BR-10, BR-11  | FR-09, FR-10        | US-10, US-11 | TS-11, TS-12, TS-13      |
| Cash-flow predictability             | BR-12         | FR-11               | US-12        | TS-14, TS-15             |
| Live reference case / customer trust | BR-13, BR-14  | FR-12               | US-13        | TS-16, TS-17             |
| Live reference case / customer trust | BR-15, BR-16  | FR-13, FR-14        | US-14, US-15 | TS-18, TS-19             |
| Scale without headcount growth       | BR-17         | FR-15               | US-16        | TS-01–TS-19 (aggregated) |
| Operator convenience                 | BR-19         | FR-16               | US-17        | TS-20                    |

## 31. Change Impact Assessment

| Dimension  | Impact                                                                                                                                                                                                                                                                                    |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| People     | Role consolidation is formalised (Section 4): one operator now explicitly holds Product Owner, Developer, Support, Account Management and Finance responsibilities. Clients gain a new self-service habit to adopt (the portal) in place of email/phone as their default contact channel. |
| Process    | Replaces ad hoc, memory-dependent manual handling end to end with the seven-process model in Section 13. Every process now has a defined trigger, flow, and — where applicable — an automated or AI-assisted decision point.                                                              |
| Technology | Introduces dependency on six external SaaS providers (Section 17) and a single relational data platform as the system of record, where none previously existed.                                                                                                                           |
| Data       | Establishes the business's first structured system of record. No historical pre-platform activity is migrated into it, with one exception: the pre-existing lead spreadsheet was migrated into the prospects table at go-live.                                                            |
| Governance | Establishes the first formal documentation pack and maintained process model for the business, together with an initial — deliberately lightweight, single-operator — RACI (Section 24) and RAID log (Section 23) where neither previously existed.                                       |

## 32. Recommendations

- Progress the Google Cloud OAuth application beyond Testing status to remove the recurring 7-day reconnection burden (addresses RISK-03).

- Formally promote the payment processor account from sandbox to live mode ahead of the first live client charge, following a documented go-live checklist (addresses RISK-04, ASM-04).
- Consider recurring or subscription invoicing for maintenance-plan clients as a Phase 2 enhancement, building on the trust established by manual invoicing plus reminders (currently Won't, Section 25).
- Consider a lightweight SLA or escalation tier on the Overview dashboard — for example, a distinct visual severity for items outstanding beyond a defined threshold — to close the residual process gap identified in Section 7.
- Plan for a second-operator or role-based access model before client volume exceeds single-operator review capacity, to mitigate the bus-factor risk documented as RISK-02.
- Periodically sample-audit auto-sent replies for accuracy as client and request volume grows, to catch any drift in AI triage behaviour early (supports NFR-01 and Decision 1/2 in Section 18).
- Revisit the daily-only scheduling constraint (RISK-05) if near-real-time inbox ingestion becomes a business priority, by moving to a higher hosting tier.